

RAWLINS, WY, USA

WMO: 726690

Lat: 41.806N

Lon: 107.200W

Elev: 2065

StdP: 78.86

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 24057

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.1	-18.1	-25.2	0.5	-21.1	-21.9	0.7	-16.9	17.3	-1.7	16.0	-1.6	2.3	60	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	32.0	13.2	30.7	12.9	29.2	12.5	15.2	25.9	14.5	25.2	13.9	24.6	5.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.3	11.5	16.8	11.1	10.6	16.4	9.8	9.7	16.6	49.2	25.6	47.2	24.9	45.1	24.6	17.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	13.8	12.2	DB	-29.7	34.3	3.8	1.2	-32.4	35.1	-34.6	35.8	-36.8	36.4	-39.5	37.3	(4)
(5)			WB	-29.7	16.6	3.5	0.7	-32.2	17.1	-34.2	17.5	-36.2	17.9	-38.7	18.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-5.2	-4.4	0.9	5.0	9.9	16.0	20.7	18.9	14.0	6.6	-0.1	-5.5
	DBStd	10.31	5.73	5.97	5.54	4.53	4.52	3.71	2.36	2.72	3.88	5.40	6.07	5.84
	HDD10.0	2302	471	404	282	158	59	4	0	0	11	127	304	482
	HDD18.3	4482	729	637	539	401	263	85	8	25	136	364	554	740
	CDD10.0	1011	0	0	1	7	55	186	333	277	131	21	1	0
	CDD18.3	148	0	0	0	0	1	16	82	43	6	0	0	0
	CDH23.3	3019	0	0	0	1	47	562	1374	819	214	3	0	0
	CDH26.7	1036	0	0	0	0	5	181	567	249	33	0	0	0
Wind	WSAvg	5.5	6.4	6.6	6.2	6.0	5.2	5.0	4.1	4.2	4.5	5.5	5.9	6.3
Precipitation	PrecAvg	246	14	13	16	28	33	23	21	21	23	24	16	15
	PrecMax	300	48	27	49	63	74	50	65	53	92	80	52	34
	PrecMin	151	1	3	6	5	1	0	1	2	3	1	1	1
	PrecStd	44	10	6	9	17	21	18	17	16	21	21	13	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	10.7	18.7	21.9	27.1	32.7	34.1	32.1	29.6	23.2	16.5	8.3
		MCWB	1.6	3.4	6.4	7.9	11.0	12.6	14.0	13.3	12.8	9.0	6.0	2.3
	2%	DB	5.3	7.4	15.3	19.1	24.5	30.7	32.6	30.6	27.5	20.5	13.7	6.0
		MCWB	0.5	1.6	5.1	6.5	9.9	12.3	13.5	12.7	11.2	8.5	4.8	1.0
	5%	DB	3.7	5.2	12.3	17.2	22.6	28.9	31.4	29.4	26.0	18.3	11.2	4.0
		MCWB	-0.2	0.6	3.8	5.8	9.3	11.7	13.3	12.5	10.7	7.5	3.7	-0.1
	10%	DB	2.4	3.4	9.6	14.8	20.2	27.0	30.1	28.0	24.0	16.0	8.8	2.5
		MCWB	-0.9	-0.3	2.7	5.1	8.5	11.1	13.0	12.2	9.9	6.5	2.6	-0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	4.1	6.8	8.6	12.5	14.6	16.1	15.9	14.1	10.6	6.9	2.9
		MCDB	6.0	9.7	17.2	19.4	23.5	27.0	26.5	25.2	25.0	19.5	14.4	7.4
	2%	WB	1.3	2.2	5.4	7.4	10.9	13.4	15.4	15.0	12.9	9.4	5.2	1.6
		MCDB	4.3	6.3	14.3	17.0	21.5	26.1	26.4	24.8	22.6	18.1	12.5	5.1
	5%	WB	0.2	1.0	4.1	6.4	10.0	12.7	14.9	14.1	11.9	8.2	4.1	0.4
		MCDB	3.2	4.6	11.3	15.7	20.1	25.2	25.8	24.0	21.5	16.9	10.5	3.7
	10%	WB	-0.8	0.0	2.9	5.4	9.1	11.8	14.3	13.4	11.1	7.0	2.9	-0.7
		MCDB	2.1	3.3	9.1	14.0	18.6	24.1	25.0	23.7	20.6	15.0	8.2	2.2
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.6	11.6	13.6	14.9	17.7	17.8	17.0	16.4	13.3	10.8	9.0
		MCDBR	9.3	10.0	15.2	18.4	19.1	21.2	20.2	19.4	19.6	16.9	13.9	9.9
		MCWBR	6.2	6.4	7.6	8.7	7.9	8.1	7.0	7.0	8.2	8.3	7.6	6.6
	5% WB	MCDBR	8.5	8.6	14.3	16.5	17.3	18.6	16.8	15.6	15.5	15.5	13.2	9.0
		MCWBR	6.0	5.7	7.4	8.2	7.5	7.3	5.7	5.7	6.4	7.7	7.4	6.3
Clear-Sky Solar Irradiance	taub	0.225	0.233	0.255	0.288	0.303	0.305	0.332	0.331	0.295	0.257	0.237	0.229	
	taud	2.511	2.518	2.527	2.459	2.473	2.475	2.430	2.421	2.521	2.616	2.580	2.519	
	Ebn at Noon	963	1002	1008	985	971	965	934	927	947	960	941	928	
	Edh at Noon	69	81	92	106	108	108	112	109	92	73	64	63	
All-Sky Solar Radiation	RadAvg	2.15	3.04	4.33	5.47	6.42	7.59	7.28	6.35	5.23	3.59	2.35	1.74	
	RadStd	0.14	0.19	0.36	0.36	0.47	0.42	0.25	0.20	0.34	0.24	0.22	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page